

# TE 456: Satellite Communication and Navigation System

## NTN – An Overview



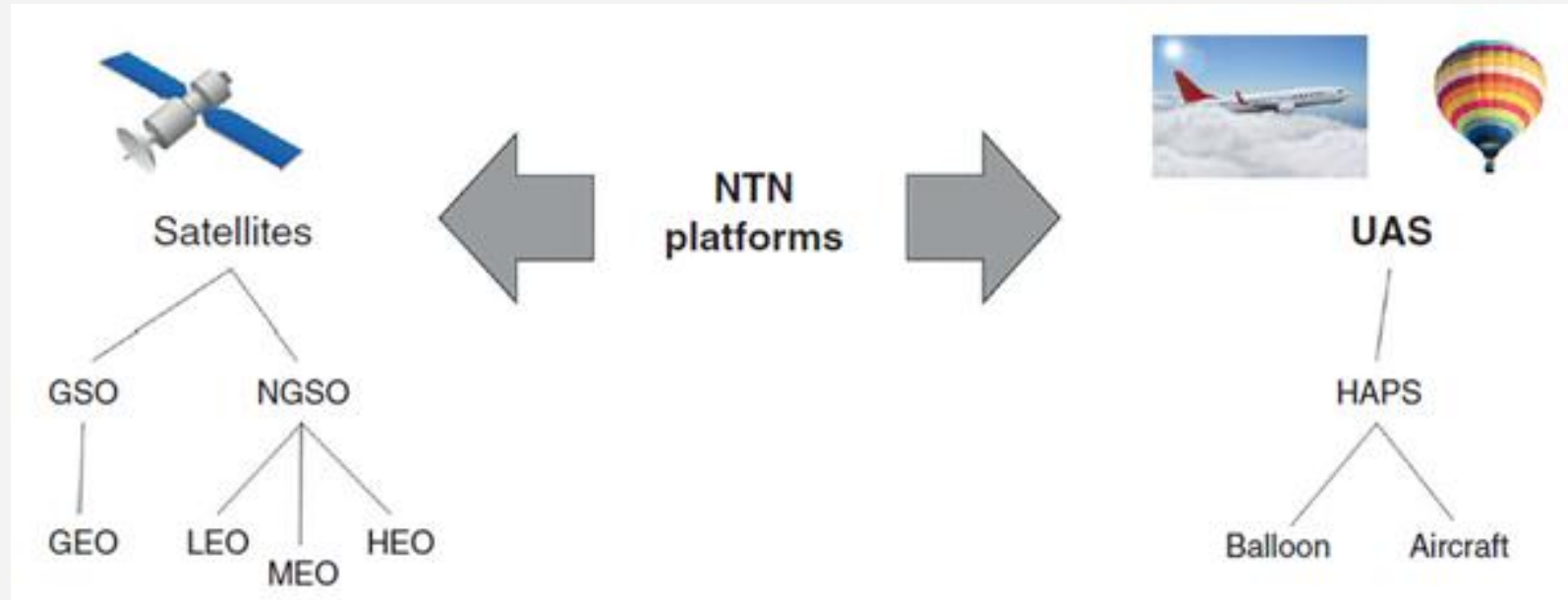
# NTN Platforms

- The NTN provides significant flexibility to operators by supporting different types of NTN platforms.
- NTN platforms include
  - Different types of satellites such as
    - Geosynchronous Earth Orbit (GEO) satellites
    - Medium Earth Orbit (MEO) satellites
    - Low Earth Orbit (LEO) satellites
  - High-Altitude Platform Stations (HAPSs)
- This unit
  - discusses the **distinct characteristics** of NTN platforms



# Types of NTN Platforms

- Spaceborne and Airborne vehicles



# Spaceborne Platform

- Spaceborne = Satellite Communication
  - Is the use of artificial satellites to provide communication links between various points on Earth
- Orbit
  - The path in which an artificial satellite travels around the Earth
    - The height of a satellite orbit
      - which is the distance of the satellite from the Earth's surface
        - determines the satellite orbital speed around the Earth



# Spaceborne Platforms

- A Satellite
  - **orbits** around the **Earth** according to **Kepler's Laws**
- Satellites (spacecraft) orbiting the Earth
  - follow the same laws that rule the motion of the planets around the sun
- Kepler's laws
  - apply quite generally to any two bodies in space, which interact through gravitation



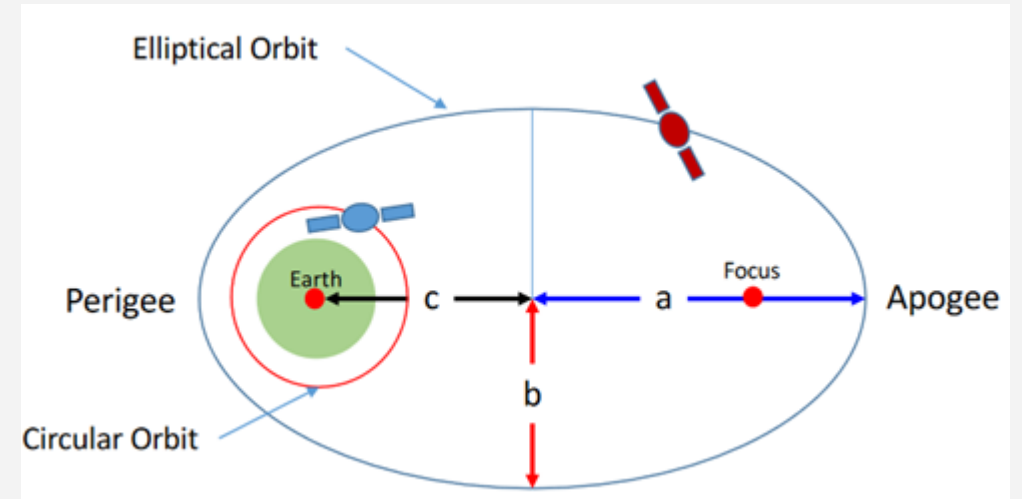


# Spaceborne Platforms

- A Satellite

- **orbits** around the **Earth** according to the **Keplerian Laws**

- Orbits are ellipses with the Earth in one of the foci
    - Getting closer to the Earth (low altitudes), the satellite's
      - speed increases
      - orbital period reduces
      - field of view reduces

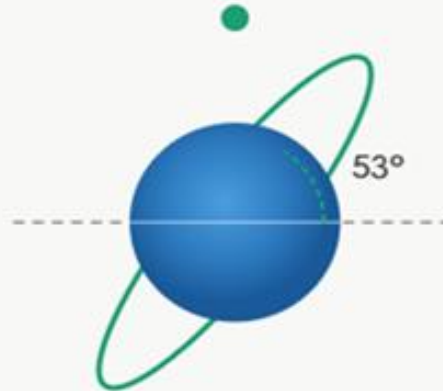


# Orbits - Inclination

Equatorial  
0° inclination

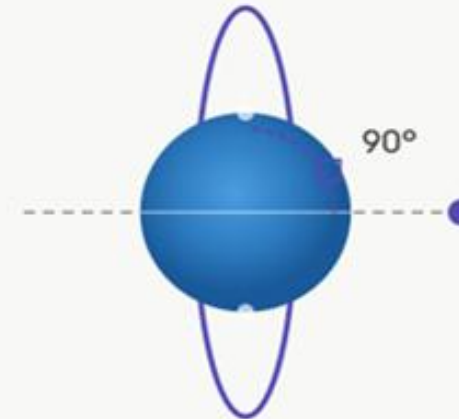


Inclined  
53° inclination



Polar  
90° inclination

...



# Period of a Satellite

- The period
  - of a satellite is the time required for a satellite to make a complete trip around the Earth
    - It is defined as a function of the distance of the satellite from the center of the Earth

$$Period = C \times distance^{1.5}$$

$$C = \frac{1}{100}$$





# Example

- The moon is located approximately 384,000 km above the Earth and the radius of the Earth is 6378 km

$$\text{Period} = (1/100) \times (384,000 + 6378)^{1.5} = 2,439,090 \text{ s} = 1 \text{ month}$$



# Example

- What is the period of a satellite that is located at an orbit approximately 35,786 km above the Earth?
  - the radius of the Earth is 6378 km

$$\text{Period} = (1/100) \times (35,786 + 6378)^{1.5} = 86,579 \text{ s} = 24 \text{ h}$$



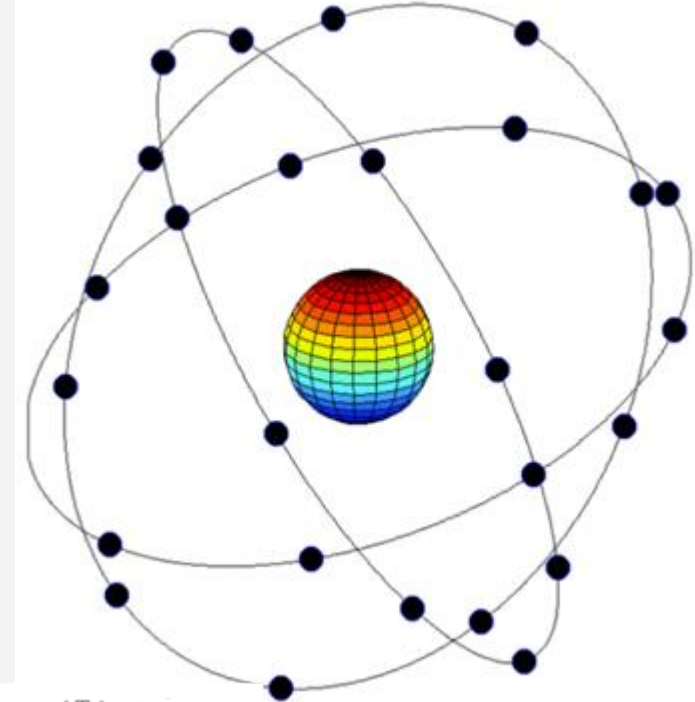
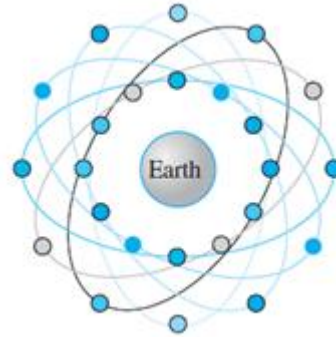
# Altitude vs Orbital Speed vs Orbital Period

| Altitude (km)             | Orbital Speed (km/s) | Orbital Period (min) |
|---------------------------|----------------------|----------------------|
| 200                       | 7.8                  | 88.3                 |
| 500                       | 7.6                  | 94.4                 |
| 1,000                     | 7.4                  | 104.9                |
| 5,000                     | 5.9                  | 201.1                |
| 10,000                    | 4.9                  | 347.4                |
| 20,200 (semi-synchronous) | 3.9                  | 718.3                |
| 35,800 (Geo-synchronous)  | 3.1                  | 1436.2 (24 hours)    |



# Constellation

- The number of satellites
  - needed for continuous and global coverage increases in low orbits
- multiple orbits (planes)
- multiple satellites per orbit (plane)



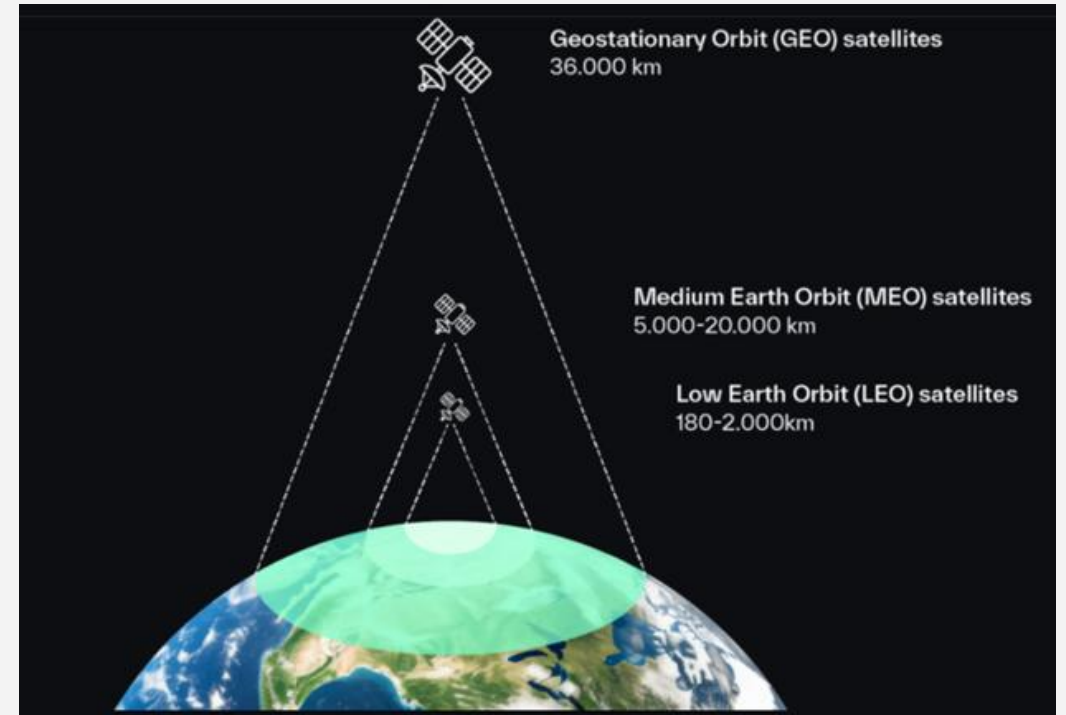
Constellations for Starlink, Kuiper and Telesat

|          | Shell | Height (km) | Orbits | Sats/orbit | inclination $\alpha$ |
|----------|-------|-------------|--------|------------|----------------------|
| Starlink | S1    | 550         | 72     | 22         | 53°                  |
|          | S2    | 1,110       | 32     | 50         | 53.8°                |
|          | S3    | 1,130       | 8      | 50         | 74°                  |
|          | S4    | 1,275       | 5      | 75         | 81°                  |
|          | S5    | 1,325       | 6      | 75         | 70°                  |
| Kuiper   | K1    | 630         | 34     | 34         | 51.9°                |
|          | K2    | 610         | 36     | 36         | 42°                  |
|          | K3    | 590         | 28     | 28         | 33°                  |
| Telesat  | T1    | 1,015       | 27     | 13         | 98.98°               |
|          | T2    | 1,325       | 40     | 33         | 50.88°               |



# Characteristics of Satellites

- The three widely used types of satellites
  - possess different characteristics and hence have different implications on the system design and supported applications or services





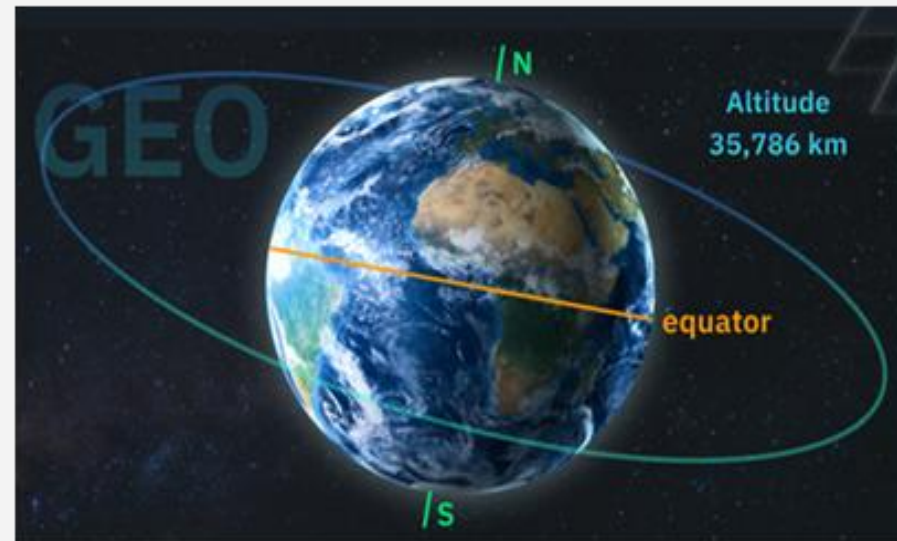
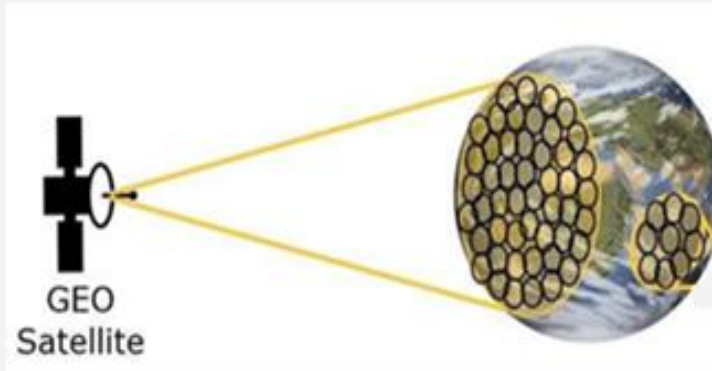
# GEO Satellites - Characteristics

- A GEO satellite
  - i. Orbits around the Earth at an altitude of 35,786 km
  - ii. Travels at the same rate as the Earth and takes 24 hours to complete one orbital period
    - As a result, it appears stationary to a fixed observer on the Earth's surface
  - iii. utilizes Earth-fixed beams
  - iv. Due to its large distance from Earth has
    - a. much longer propagation delay between the user device and the satellite (RTT ~ 600ms)
    - b. large propagation path loss



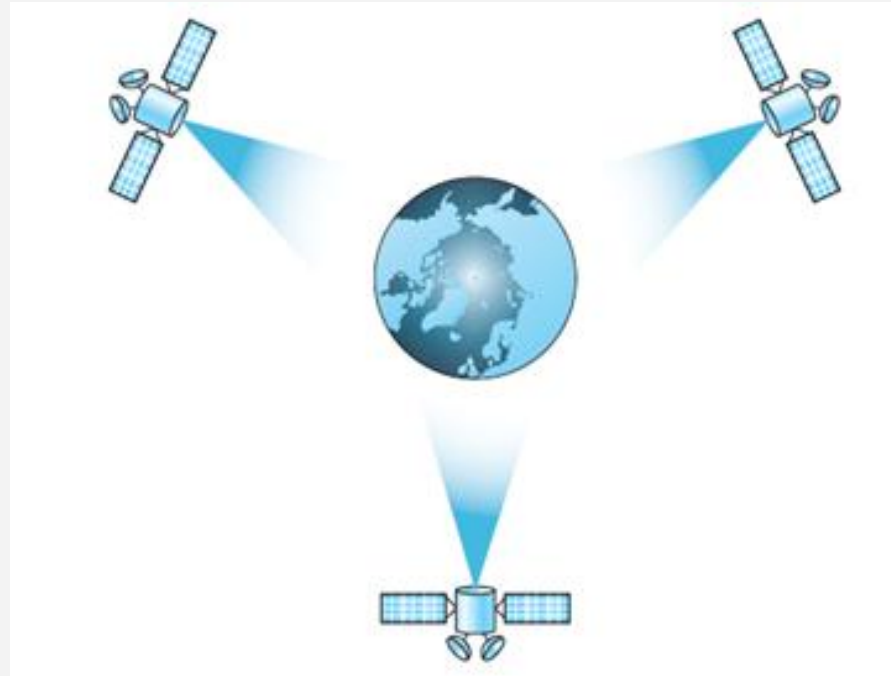
# GEO Satellites - Characteristics

- GEO
  - Any satellite orbit that is fixed at the 35,786 km equatorial line



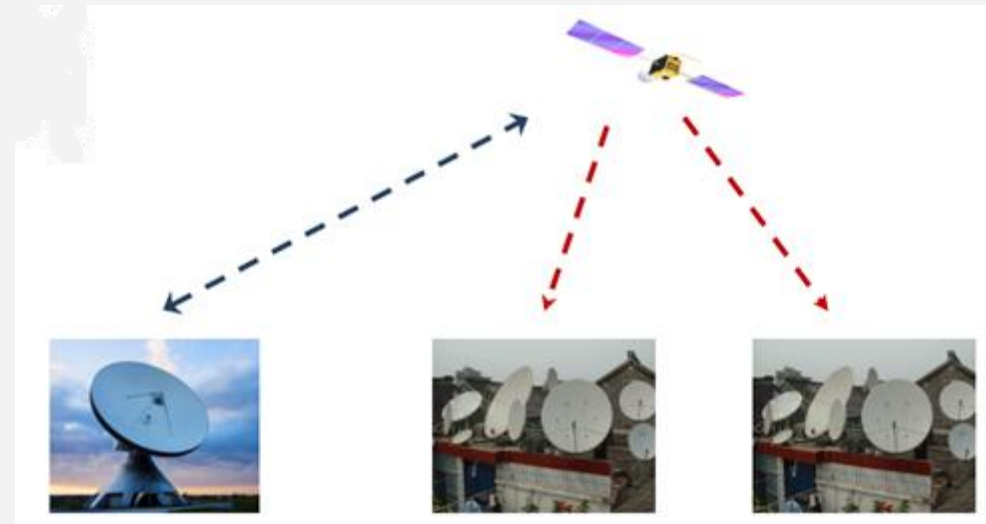
# GEO Satellites - Characteristics

- Due to the high altitude
  - GEO satellites can cover a large portion of the Earth's surface
    - Theoretically, three GEO satellites can provide near-global coverage



# GEO Satellites - Characteristics

- The beam footprint of GEO satellites
  - can range from 200 kilometers to 3,500 kilometers
- There are currently hundreds of GEO satellites in orbit,
  - and most of them are used to provide services such as
    - a. weather monitoring
    - b. TV broadcasting
    - c. remote sensing and positioning



# GEO Advantages & Disadvantages

- Pros:

- Simple and cost-effective ground tracking (stationary antennas)
- Massive, uninterrupted area coverage per satellite
- No need for satellite tracking or handovers on the ground antenna

- Cons:

- High Latency: High altitude means a longer signal round-trip (~ 600ms)
- Poor signal coverage at the extreme polar regions





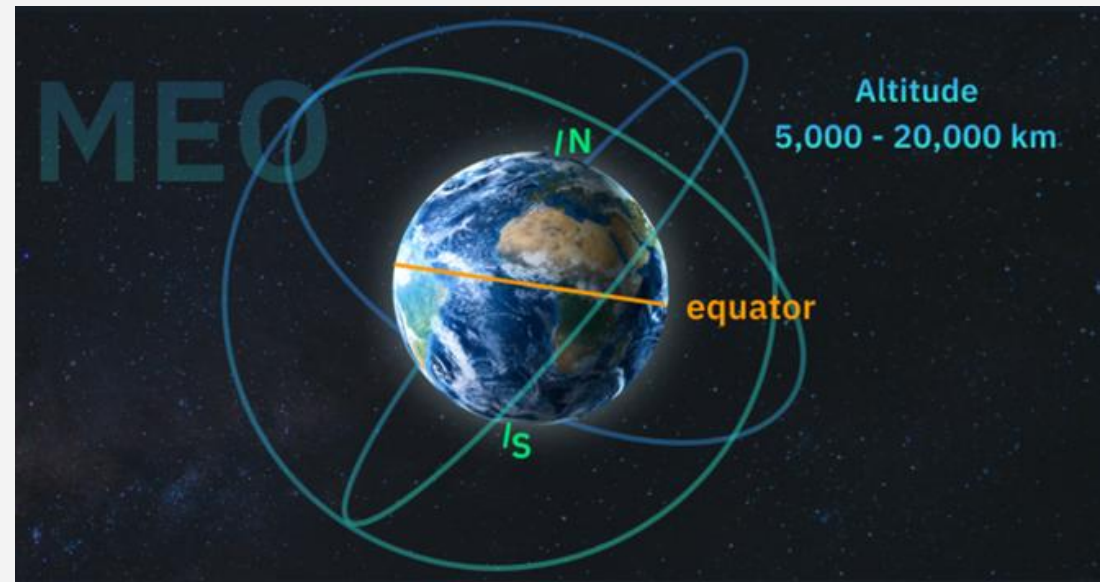
# Non-Geostationary Earth Orbit (NGEO)

- NGEO satellites
  - operate at lower altitudes with orbital periods of less than 24 hours
    - In addition, their positions can always be changed with respect to observers on Earth
- There are two types of NGEO satellites according to their altitudes:
  - i. Medium Earth Orbit (MEO) satellites
  - ii. Low Earth Orbit (LEO) satellites



# MEO Satellites Characteristics

- They perfectly
  - balance the wide coverage of high orbits and the low-latency signal speeds of low orbits
    - Because of this, they are the backbone of global navigation and high-speed enterprise connectivity



# MEO Satellites Characteristics

- A MEO Satellite
  - i. Orbits at altitudes ranging from 7,000 to 25,000 km
  - ii. travels at about 3.83 km/s or 13,800 km per hour
  - iii. Has an orbital period of 6 to 12 hours
  - iv. Number of satellites for full Earth coverage, 6
  - v. uses quasi-Earth-fixed beams or Earth-moving beams
  - vi. Can create beams with diameters from 100km to 500km
  - vii. Based on its distance from Earth, MEO has
    - medium propagation delay (23.3 to 83.2ms)
    - medium propagation path loss



# LEO Satellites Characteristics

## ■ A LEO Satellite

- i. Orbits orbit at altitudes ranging from 300 to 1,500 km
- ii. travels at about 7.78 km/s or 28,000 km per hour
- iii. Has an orbital period of 90min
- iv. Number of satellites for full Earth coverage, hundreds
- v. uses quasi-Earth-fixed beams or Earth-moving beams
- vi. Can create beam footprints with diameters from 5km to 200km
- vii. Based on its distance from Earth, LEO has
  - small propagation delay (1 to 5ms)
  - small propagation path loss



# NGEO Advantages & Disadvantages

## ■ Pros:

- Low Latency: Closer to Earth, delivering quick response times (~ 20-40ms)
- Better signal penetration and coverage over the polar regions
- High-speed broadband delivery to remote or polar regions

## ■ Cons:

- Requires dynamic "hand-offs" between satellites as they pass overhead
- Ground antennas must actively track moving satellites and perform "handovers"
- Requires a large number of satellites (mega-constellations) to achieve global coverage





# NGEO -Applications

- MEO and LEO satellites
  - can be used for sensing, positioning, and communication systems due to their low propagation delay compared to GEO satellites
- For example,
  - The U.S. Global Positioning System (GPS) uses at least 24 MEO satellites for its global positioning services
- Although MEO satellites
  - can be used for communication services, the communication latency is still high and may not be feasible for today and future communication applications



# NGEO -Applications

- LEO satellites
  - Have emerged as promising platforms to provide Internet services to users on Earth, anywhere and anytime
- For example,
  - Starlink, the world's first and largest constellation, uses thousands of LEO satellites
    - to deliver broadband Internet for services such as streaming, online gaming, and video calls to users around the world
- Starlink's satellites
  - fly at an altitude of about 550 kilometers with much
    - lower propagation delay compared to those of GEO and MEO satellites.
      - As a result, Starlink can achieve a round-trip delay of around 25 milliseconds for its services

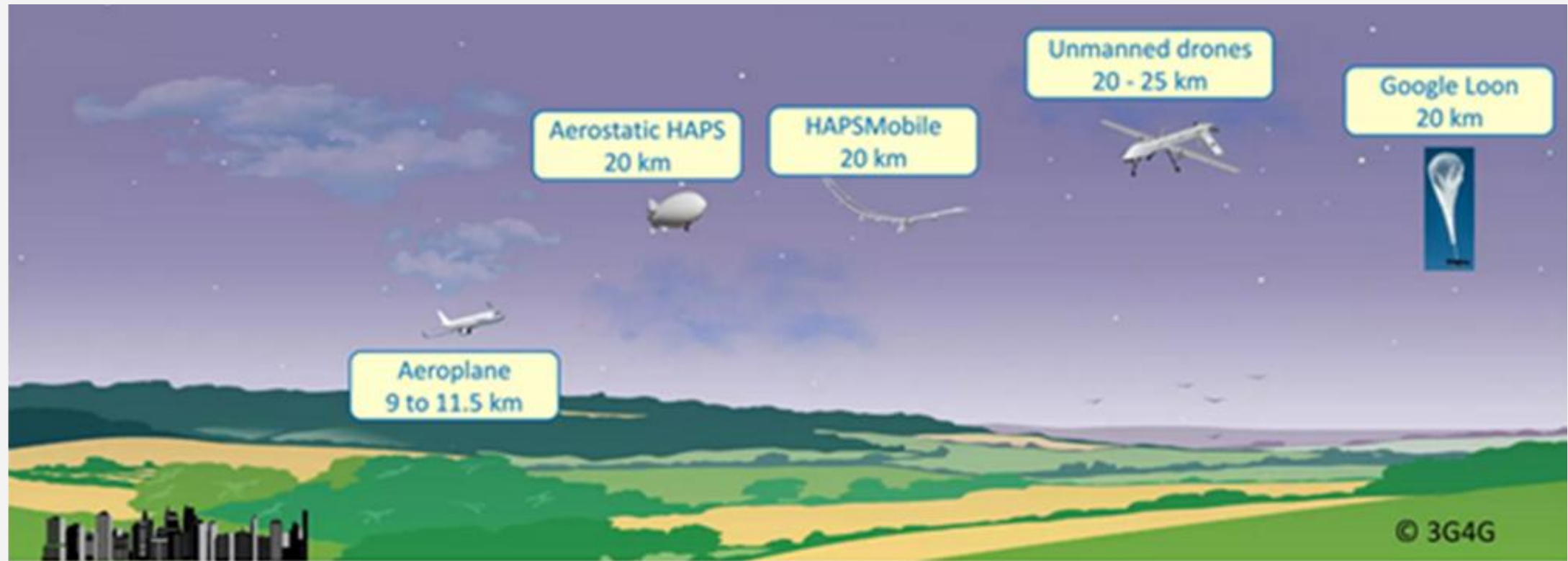


# NGEO -Applications

- The current biggest competitor of Starlink is
  - Amazon's Project Kuiper, which aims to launch over 3,236 LEO satellites
    - to provide low latency and high-speed Internet services to users on a global scale



# Airborne Platforms



# Airborne Platforms

- Airborne NTN

- can be deployed quickly at a lower cost and has a much smaller propagation delay compared to spaceborne NTN
  - However, airborne platforms face critical challenges of stabilization in the air and refueling
    - In particular, operating at low altitudes makes them vulnerable
      - To environmental conditions such as strong winds or storms that can change the flying path of balloons and drones or even destroy them

- While airborne platforms

- may not be effectively used to provide Internet services
  - they can be used in emergency scenarios such as during natural disasters and rescue missions in rural areas where cellular coverage is not available





# Airborne Platforms - Characteristics

## ■ The HAPS

- The 3GPP focuses on the altitude between 8 and 50 km
- maintains its position relative to the Earth's surface and hence appears as stationary to a fixed observer on the Earth's surface
- uses Earth-fixed beams like a GEO satellites
- propagation delay can be considered comparable to TNs
- propagation path loss is much smaller than that for LEO systems
- coverage can be viewed as on-demand coverage
- typically covers a larger area compared to a TN gNB and a smaller area than a satellite

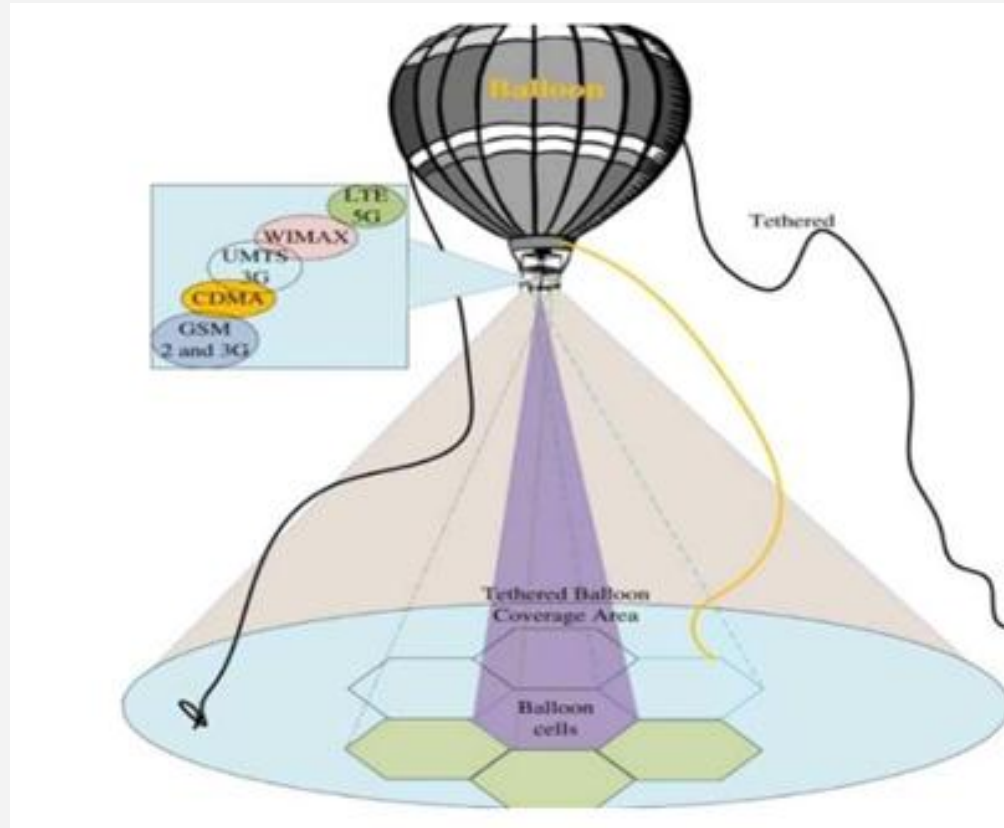


# Airborne Platforms

- Project Loon
  - from Google deploys balloons at high altitudes from 18 kilometers to 25 kilometers to provide connectivity to remote and rural areas
- Project Aquila
  - From Facebook developed solar-powered drones that operate at an altitude of up to 27 kilometers and provide Internet services to an 80-kilometer-radius area below its flying path



# Airborne Platforms



# NTN - Characteristics

| Platforms                                 | Altitude range             | Orbit                                                                                                     | Typical beam footprint size |
|-------------------------------------------|----------------------------|-----------------------------------------------------------------------------------------------------------|-----------------------------|
| Low-Earth Orbit (LEO) satellite           | 300 – 1500 km              | Circular around the earth                                                                                 | 100 – 500 km                |
| Medium-Earth Orbit (MEO) satellite        | 7000 – 25000 km            |                                                                                                           | 100 – 500 km                |
| Geostationary Earth Orbit (GEO) satellite | 35 786 km                  | notional station keeping position fixed in terms of elevation/azimuth with respect to a given earth point | 200 – 1000 km               |
| UAS platform (including HAPS)             | 8 – 50 km (20 km for HAPS) |                                                                                                           | 5 - 200 km                  |
| High Elliptical Orbit (HEO) satellite     | 400 – 50000 km             | Elliptical around the earth                                                                               | 200 – 1000 km               |





| Platform                                   | Features                                                                                                                                                                                                                                                                                                                                               | Main Use Cases                                                                                                                          |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Low Earth Orbit (LEO) Satellite            | Altitude 300 to 1500 km<br>Ground coverage ranges from 100 to 1000 km.<br>Circular orbit around Earth<br>Low latency<br>The coverage area per satellite is narrower than that of other satellites.<br>Many satellites are required for coverage.                                                                                                       | Connections to mountainous and rural areas, etc.<br>International Space Station (ISS), Hubble Space Telescope, telecommunications, etc. |
| Medium Earth Orbit (MEO) satellites        | Altitude 7000 to 25000 km<br>Ground coverage ranges from 100 to 1000 km.<br>Circular orbit around Earth<br>Latency is high due to the higher altitude compared to LEO.<br>Fewer satellites are needed to provide coverage than in LEO.                                                                                                                 | Global Navigation Satellite Systems (GNSS) include GPS, GLONASS, Galileo, communications, remote sensing, etc.                          |
| Geostationary Earth Orbit (GEO) satellites | Located about 36,000 km above the equator<br>Ground coverage ranges from 200 to 3500 km.<br>Circular orbit with an inclination of 0 degrees above the equator<br>Satellite orbital period coincides with Earth's rotation.<br>The satellite is fixed over the same point.                                                                              | Communication, broadcasting, weather observation, etc.                                                                                  |
| Geosynchronous Orbit (GSO)                 | Altitude approx. 36,000 km<br>Ground coverage ranges from 200 to 3500 km.<br>Circular or elliptical orbit around Earth<br>The orbital period of the satellite coincides with Earth's rotational period.<br>No restrictions on orbit inclination or orbit eccentricity<br>Fixed in the vicinity of a specific point in the sky<br>GEO is a type of GSO. | Communications, positioning services such as Quasi-Zenith Satellite MICHIBIKI                                                           |
| High Altitude Platform Station (HAPS)      | Operates in the stratosphere at altitudes of 20 to 50 km.<br>Ground coverage ranges from 2 to 200 km.<br>Unmanned platform, Lower latency than LEO<br>Highly maintainable compared to satellites<br>Including balloons and airships                                                                                                                    | Communication, surveillance, environmental monitoring, etc.                                                                             |
| Unmanned Aircraft System (UAS)             | Unmanned Aerial Vehicles (UAVs) and drones<br>Drone altitude is limited to 90 to 150 m, depending on country, region, flight area, etc.                                                                                                                                                                                                                | Communication, monitoring, delivery, etc.<br>Rapidly deployable in areas hit by natural disasters, etc.                                 |





# Types of Beams

- Three types of beams can be used in an NTN to provide radio coverage:
  1. Earth-fixed beams
  2. Quasi-Earth-fixed beams
  3. Earthmoving beams
- These beam types are also referred
  - to as **types of NTN cells** or **types of service links**

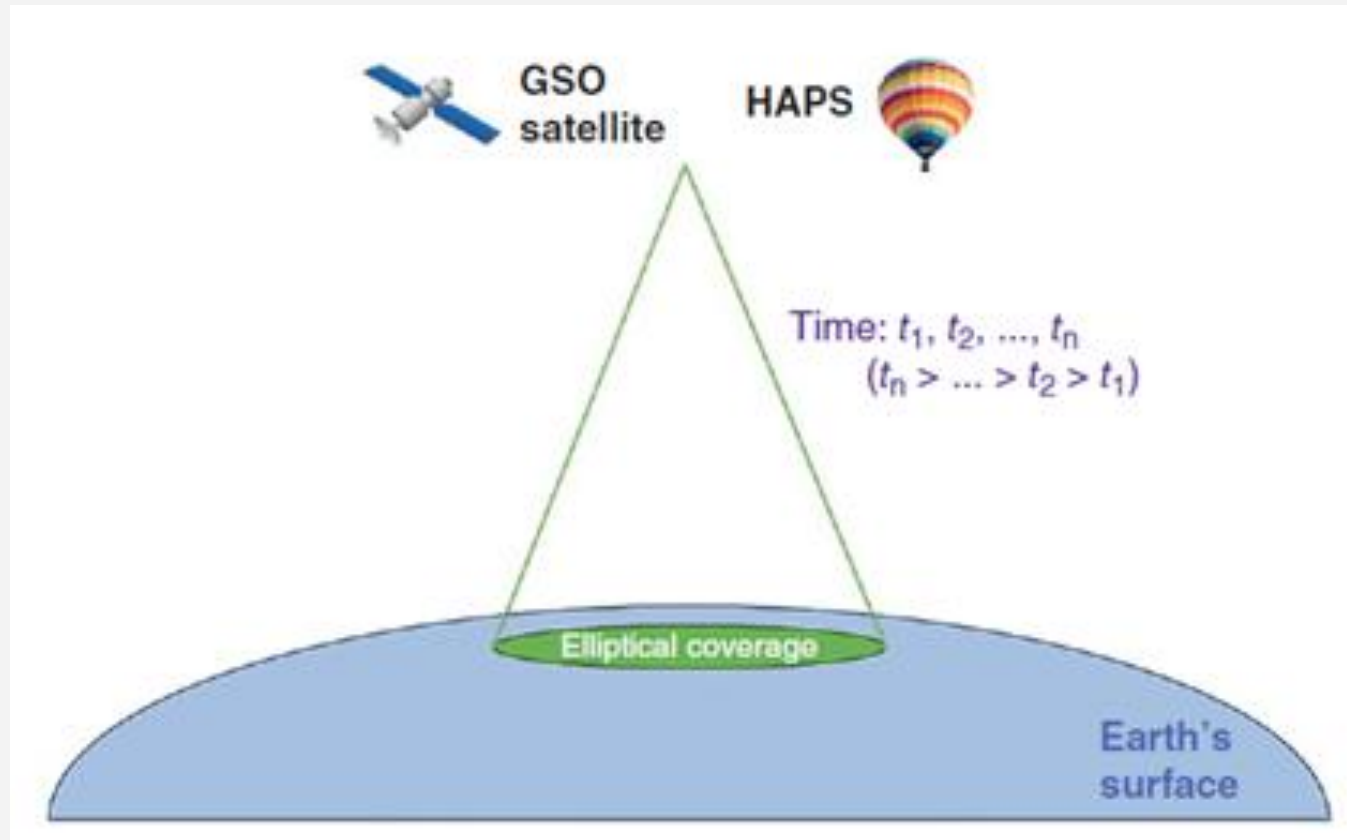


# Earth-Fixed Beams

- An Earth-fixed beam
  - covers a fixed region on the Earth's surface at all times
    - For example, the NTN platform, such as a GEO satellite or a HAPS
      - provides radio coverage to the same geographic area at all time instants  $t_1, t_2, \dots, t_n$  and so on
        - These platforms are stationary in space or air and create fixed beams toward the Earth's surface



# Earth-Fixed Beams

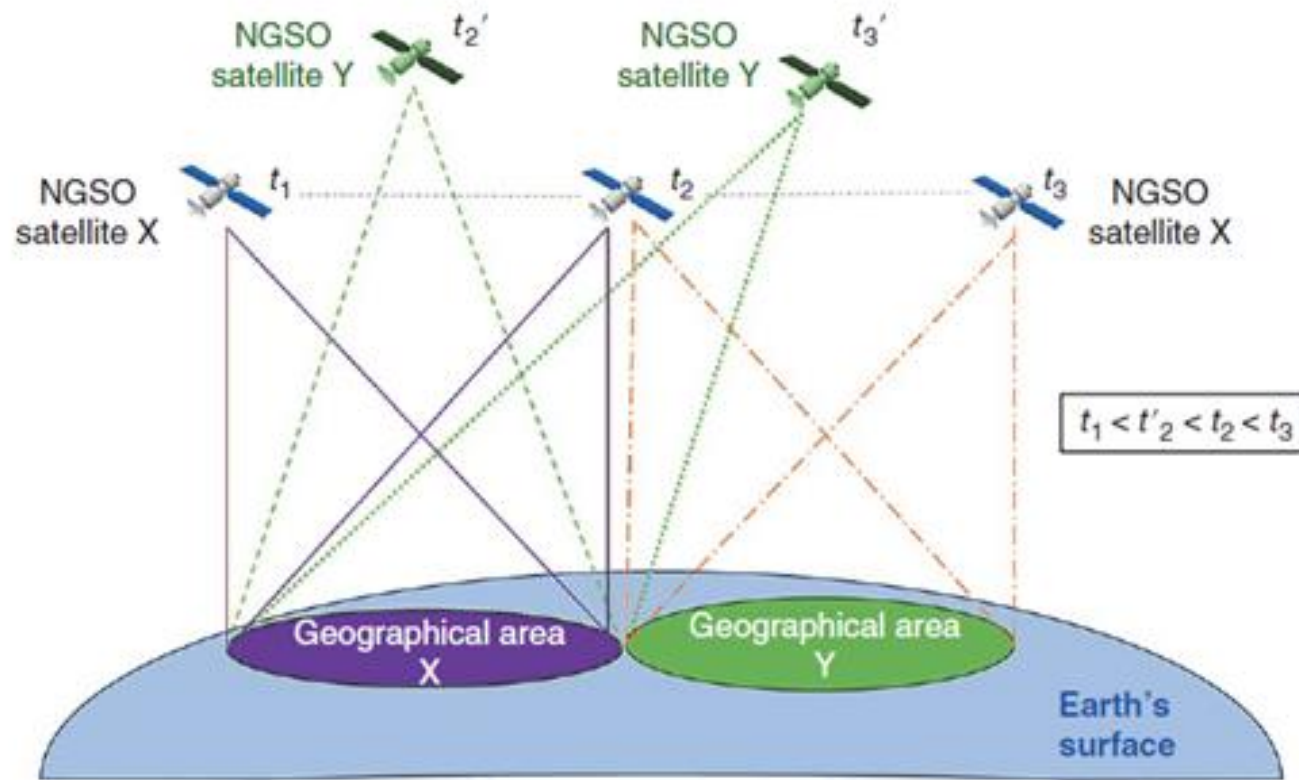


# Quasi-Earth-fixed beams

- A quasi-Earth-fixed beam
  - covers a given geographic area X during the period from  $t_1$  to  $t_2$ 
    - Then, such beam covers a different geographic area Y during the period from  $t_2$  to  $t_3$ 
      - During the period from  $t_2$  to  $t_3$ , a different quasi-Earth-fixed beam covers the geographic area X
- The time overlap
  - between the outgoing and incoming beams ( $t_2'$  to  $t_2$ ) enables the transfer of user devices from one beam to the other



# Quasi-Earth-fixed beams





# Quasi-Earth-fixed beams

- NTN platforms such as NGSO ( MEO and LEO) satellites
  - may have beam steering capability such that the satellites can continue to illuminate the same geographic area even as the satellites move from one location to another in their orbits
- In summary
  - to provide coverage continuously in a given geographic area, an incoming quasi-Earth-fixed beam replaces an outgoing quasi-Earth-fixed beam

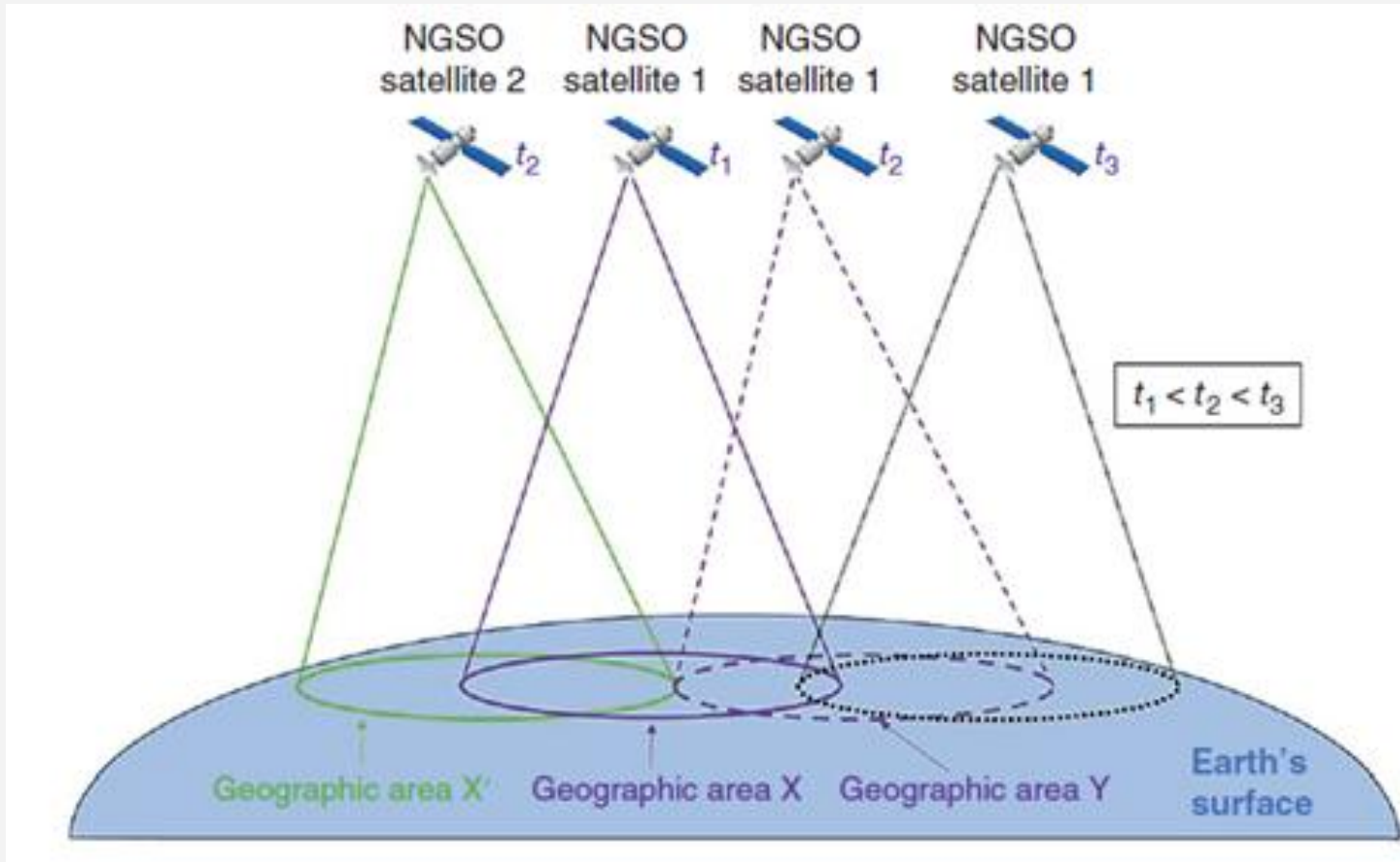


# Earth-Moving Beams

- An Earth-moving beam
  - covers a given geographic area X at the instant  $t_1$ 
    - Then, such a beam covers a slightly different geographic area Y at the very next instant  $t_2$ 
      - The beam covers a yet another geographic area Z at the next instant  $t_3$- NTN platforms such as NGSO (MEO and LEO) satellites
  - may use fixed beams, and the movement of the NTN platform leads to different geographic areas being illuminated at different instants



# Earth-Moving Beams



# Earth-Moving Beams

- While one Earth-moving beam
  - ceases to cover a geographic area, another Earth-moving beam starts covering such a geographic area
- For example,
  - at the instant  $t_2$ , when NGSO satellite 1 covers the geographic area Y, NGSO satellite 2 covers geographic area X' to provide continuous and contiguous coverage
- In summary,
  - to provide coverage continuously in a given geographic area, an incoming Earth-moving beam replaces an outgoing Earth-moving beam





# THANK YOU

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